

Next-generation Bayesian local robustness

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Berkeley Neyman Seminar

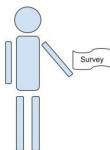
September 2026

The basic problem

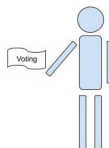
We have a survey population, for whom we observe:

- Covariates x (e.g. race, gender, zip code, age, education level)
- Responses y (e.g. A binary response to “do you support candidate Z”)

We want the average response in a target population, in which we observe only covariates.



Observe (x_i, y_i) for $i = 1, \dots, N_S$



Observe x_j for $j = 1, \dots, N_T$

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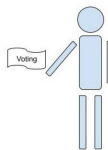
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The problem is that the populations may be very different, maybe leading to bias.



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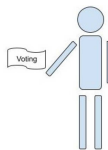
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This talk will study “Multilevel regression and poststratification” estimators:

- Set $\mathbb{E}[y|x, \theta] = m(\theta^\top x)$ and prior $\mathcal{P}(\theta)$ (e.g. Hierarchical logistic regression)
- Estimate the posterior $\mathcal{P}(\theta|\text{Survey data})$ with Markov Chain Monte Carlo (MCMC)
- Take $\hat{\mu}^{\text{MrP}}(Y_S) = \frac{1}{N_T} \sum_{j=1}^{N_T} \hat{y}_j$ where $\hat{y}_j = \mathbb{E}_{\mathcal{P}(\theta|\text{Survey data})}[y|x_j]$.

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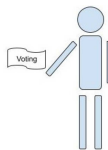
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Problem: MCMC is time consuming and the regression model is surely misspecified.

Goal: Meaningful regression specification checks with frequentist uncertainty.

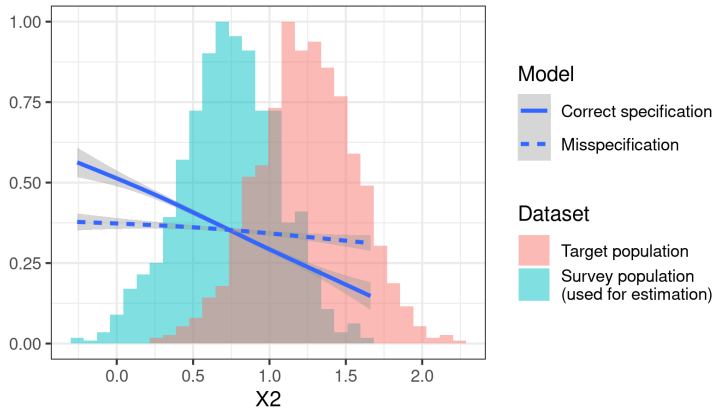


Figure 1: A covariate shift diagnostic

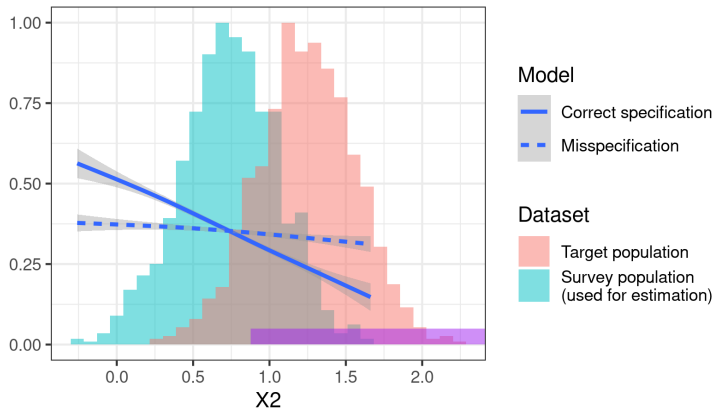


Figure 1: A covariate shift diagnostic

- Meaningful Bayesian local specification diagnostics
- Frequentist variability of Bayesian posteriors with the infinitesimal jackknife
- Flow-based Bayesian local robustness

Result preview

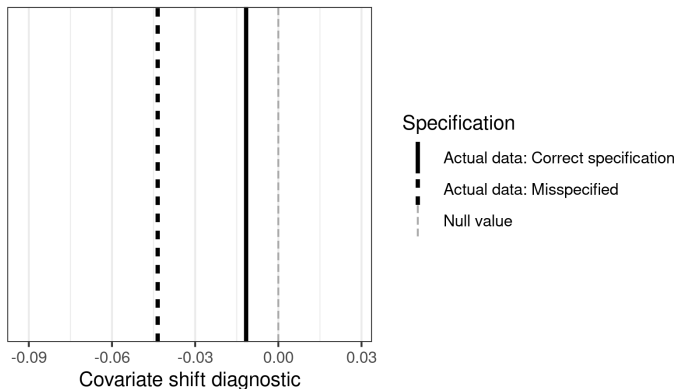


Figure 2: A covariate shift diagnostic

Here is the check for two models on the same simulated data: one correctly specified and one misspecified. The misspecified model is more extreme.

Problem: What is “large enough” to be real evidence of misspecification?

Possible answer: Frequentist confidence intervals via the parametric bootstrap.

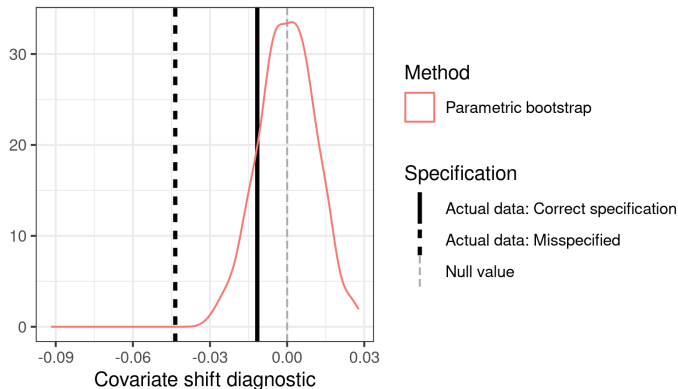


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The frequentist interval based on the parametric bootstrap rejects the misspecified model but not the correctly specified model.

Problem: Expensive: each bootstrap sample required a new MCMC run.

Possible answer: Infinitesimal jackknife confidence intervals?

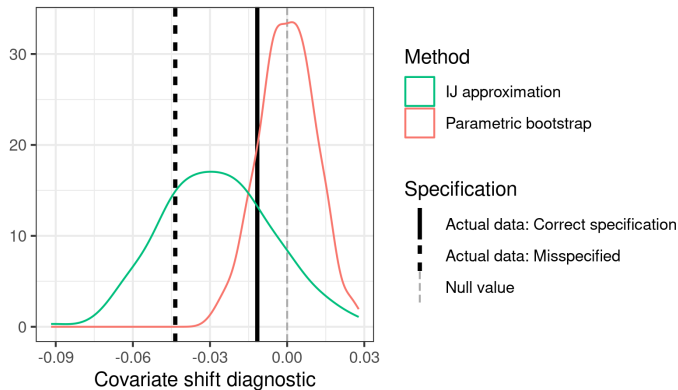


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IJ confidence intervals are fast and often work well for posterior means but are not accurate for this more complex diagnostic (due in part to Monte Carlo error).

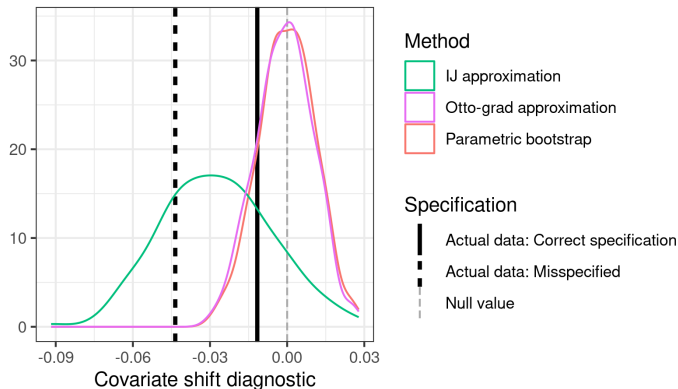


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Answer: Flow-based local sensitivity measures (“Otto grad”) produce good approximations to the bootstrap at negligible computational expense.

